Lab 3.2

Single View Measurements

09/14/26

Jude Shin

1) $f = 3,070.5$

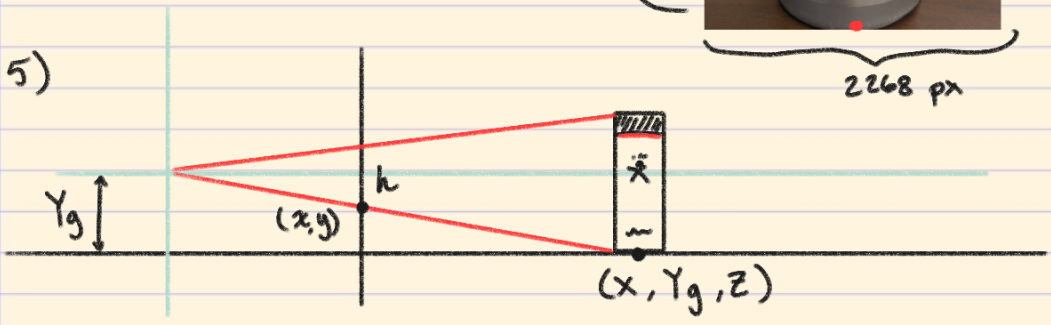
2)

3) $Z_r = 75$ inches (real)
$Y_g = 5.25$ inches
$H_r = 10$ inches (real)

4) $h = 4032$ px
$y = 4032$ px




5)




* $Y = Y_g \Rightarrow y = f\frac{Y_g}{Z} + c_y \Rightarrow Z = f\frac{Y_g}{y - c_y}$

$Z = 3070.5$ px $\frac{5.25 \text{ in}}{4032 \text{ px} - 2016 \text{ px}}$

$= \frac{3070.5 \text{ px}}{2016 \text{ px}} \cdot 5.25$ in

$= \boxed{7.996 \text{ in}}$ Estimated distance from bottle to camera

* $H = h\frac{Z}{f}$ $H = 4032 \text{ px} \frac{7.996 \text{ in}}{3070.5 \text{ px}}$

$H = \boxed{10.4998 \text{ in}}$ Estimated water bottle height

COMPARISON:

depth: expected: 7.5 in estimated: 7.996

variance: 6.61 %

height: expected: 10 in estimated: 10.4998

variance: 5.00 %